AIS — Lab reference guide

# Key Concepts: Active Directory + GLPI

Site A project — Windows Server 2022 (AD DS, DNS) · Debian 13 · GLPI 11 · LDAP

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## 1. The big picture

A company needs two things for its IT users: **one identity per person** and **one place to ask for help**.

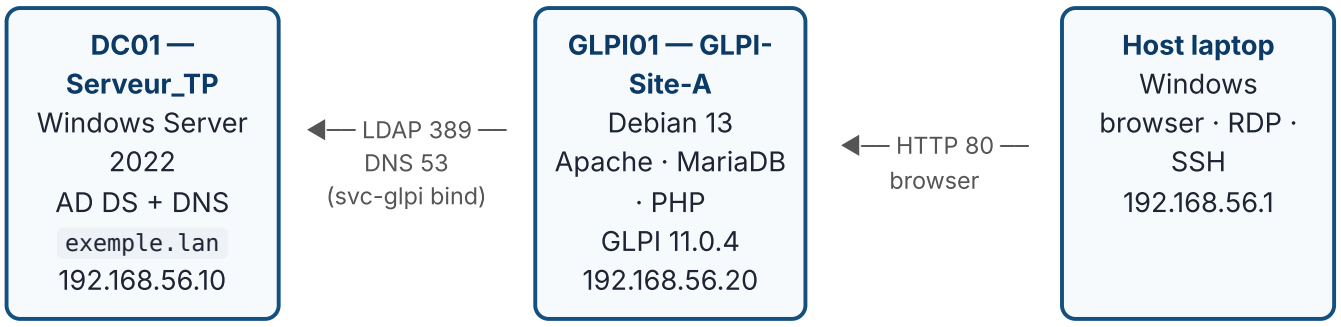
- **Active Directory (AD)** is the identity system. It stores who the users are, which groups they belong to, and checks their passwords.
- **GLPI** is the IT service-management tool: help-desk tickets plus the inventory of computers and network devices.
- **LDAP** is the language that lets GLPI ask AD "does this user exist and is this password right?". Users log in to GLPI with their normal Windows account; no second password is needed.

> **Analogy:** AD is the company's HR register and badge office. GLPI is the help desk at reception. LDAP is the internal phone line the receptionist uses to call HR and check a badge before helping someone.

The project follows a real "put into production" workflow:

1 Inventory → 2 Network & DNS → 3 AD structure → 4 LDAP link → 5 Profiles → 6 Asset inventory → 7 Ticket → 8 Acceptance tests

## 2. Lab architecture and IP plan




| | Home lab | School design (target) |
|---|---|---|
| Server network | VirtualBox host-only 192.168.56.0/24 (flat) | VLAN 30 – Server 172.16.0.64/27 |
| Network devices | None (declared manually in GLPI) | SW1 172.16.0.129, R1 172.16.0.162, FW 172.16.0.174 |
| Internet for GLPI | Second adapter in NAT mode | Through the router and firewall |

> **Rule to remember:** two machines talk directly only if they are on the **same subnet**. Otherwise a **router** is needed. At the start AD (192.168.20.x) and GLPI (192.168.56.x) were on different subnets with no router, so nothing could work until GLPI was moved to 192.168.56.20.

## 3. Networking basics that made or broke the lab

| Concept | Plain meaning | Where it mattered |
|---|---|---|
| IP address / mask | The address of a machine and the size of its neighbourhood (/24 = 256 addresses, same first three numbers). | Putting AD and GLPI in 192.168.56.0/24. |
| Gateway | The "exit door" to other networks. | GLPI uses the NAT adapter's gateway for internet access; the host-only adapter has none. |
| VirtualBox **NAT** | The VM reaches the internet through the host, but nobody can reach the VM. | GLPI's internet access (apt install). |
| VirtualBox **Host-only** | A private network between the host and its VMs. | The lab network 192.168.56.0/24. |
| "Not attached" / cable connected | A virtual network card can exist with no cable plugged in. | The first fault found in Stage 1. |
| Snapshot | A restore point of a VM. | Taken before every change. |
| Full clone | An independent copy of a VM, with new MAC addresses. | GLPI-Site-A was cloned from debian-tp so another class's VM stayed untouched. |
| Port | A numbered door for a service on a machine. | 22 SSH, 53 DNS, 80 HTTP, 389 LDAP, 636 LDAPS, 3389 RDP. |

### Debian network files

- `/etc/network/interfaces` : static IP configuration (ifupdown). Apply it with `ifdown/ifup <iface>` .
- `/etc/resolv.conf` : which DNS server to ask. It can be overwritten by DHCP clients; in this lab **dhcpcd** kept replacing it, which was fixed with `nohook resolv.conf` in `/etc/dhcpcd.conf` .
- `/etc/hosts` and `hostnamectl` : the machine's own name (glpi01).

## 4. DNS — the phone book of the domain

**DNS (Domain Name System)** turns names into IP addresses: `glpi01.exemple.lan → 192.168.56.20` . Active Directory **depends on DNS**. Clients find the domain controller through DNS records, so if DNS is wrong, AD looks "broken" even when it is fine.

| Record | Meaning | Example |
|---|---|---|
| A | Name → IPv4 address | glpi01 → 192.168.56.20 |
| PTR (reverse zone) | IP address → name | 192.168.56.20 → glpi01.exemple.lan |
| SRV | "Where is the service?" (used to locate the DCs) | _ldap._tcp.exemple.lan |
| Zone | The set of records for one domain | exemple.lan |

> **Lessons from the lab:**
> • A DC must use **itself** as its DNS server, not 8.8.8.8, or it cannot find its own domain.
> • An old network card left enabled had re-registered a stale address (192.168.20.10), so `nslookup` returned two answers. Disable the unused card, then delete the stale records.
> • Always test both ways: name → IP and IP → name.

## 5. Active Directory

| Term | Plain meaning | In this project |
|---|---|---|
| **Domain** | A security boundary sharing one user database. | `exemple.lan` |
| **AD DS** | The Windows role that provides the directory. | Installed on Serveur_TP |
| **DC** (Domain Controller) | The server holding the directory and checking logins. | DC01 / WIN-RG5VDD2MS10 |
| **OU** (Organizational Unit) | A folder for objects. It lets you apply a GPO or delegate admin rights to one part of the directory. | Utilisateurs (Direction, Employes, IT), Groupes, Postes, ComptesService |
| **Security group** (global) | A list of users who get the same rights. Grant rights to groups, never to individual users. | GRP-Direction, GRP-Employes, GRP-IT, GRP-GLPI-SVC |
| **User account** | A real person. | direction01, employe01, it01 |

| | | |
|---|---|---|
| **Service account** | An account used by an application rather than a person. Minimum rights, no interactive login. | svc-glpi (only reads the directory) |
| **GPO** | A Group Policy Object: settings pushed to the users and computers in an OU. | Reason for separating OUs |

```
DC=exemple,DC=lan
├── OU=Utilisateurs
│   ├── OU=Direction        direction01  → GRP-Direction
│   ├── OU=Employes         employe01    → GRP-Employes
│   └── OU=IT               it01         → GRP-IT
├── OU=Groupes              GRP-*
├── OU=Postes               (computers)
└── OU=ComptesService       svc-glpi     → GRP-GLPI-SVC
```

> **Why separate OUs?** Different objects need different rules (GPOs, delegation, monitoring). Separate OUs also let GLPI search only `OU=Utilisateurs`, so service accounts are never imported as people.

## 6. LDAP — how GLPI talks to Active Directory

**LDAP (Lightweight Directory Access Protocol)** is the standard way to **search** a directory and **check a password** against it. AD answers LDAP on port **389** (unencrypted) and **636** (LDAPS, encrypted with TLS).

> **How a GLPI login works:**
> 1. GLPI connects to the DC as `svc-glpi` (the **bind**).
> 2. It **searches** under the Base DN for a user whose `sAMAccountName` = what was typed.
> 3. It re-binds as that user with the typed password. If AD accepts, the login succeeds.
> 4. It identifies the person by `objectGUID` so their tickets and assets stay linked to them.

| GLPI setting | Meaning | Lab value |
|---|---|---|
| Server / Port | Where the DC listens | 192.168.56.10 / 389 |
| **DN** (Distinguished Name) | The full "address" of an object in the tree, read right to left | CN=svc-glpi,OU=ComptesService,DC=exemple,DC=lan |
| **Base DN** | Where in the tree to start searching | DC=exemple,DC=lan |
| **Bind DN** | The account GLPI uses to connect and search | svc-glpi (read only, not an admin) |
| Login field | The attribute matched against the typed username | samaccountname |
| Sync field | A stable unique ID | objectguid |
| Filter | Which objects count as "users" | `(&(objectClass=user)(objectCategory=person))` |

### Key attributes

- **sAMAccountName**: the short login name (it01). It can change.
- **objectGUID**: a unique ID that **never changes**, even after a rename or a move to another OU. That is why it is used for synchronisation.
- **CN / OU / DC** in a DN: Common Name / Organizational Unit / Domain Component.
- **Referrals**: AD can answer "ask another server". GLPI usually has referrals turned off for a single DC.

### Testing LDAP outside GLPI (from Debian)

```
timeout 3 bash -c '</dev/tcp/192.168.56.10/389' && echo "Port 389 OPEN"
ldapsearch -x -H ldap://192.168.56.10 -D "CN=svc-glpi,OU=ComptesService,DC=exemple,DC=lan" -
W \
  -b "DC=exemple,DC=lan" "(sAMAccountName=it01)" dn
# -W prompts for the password: it is never written in the command or history
```

"result: 0 Success" plus the user's DN shows that network, DNS, the account and the Base DN are all correct, so any remaining problem is in the GLPI settings.

## 7. GLPI

**GLPI** (Gestionnaire Libre de Parc Informatique, "free IT asset manager") is a web application (PHP + MariaDB) for **ITSM**: help desk, asset inventory, contracts and knowledge base. In GLPI 11 the URL is the server root (`http://192.168.56.20/`), not `/glpi`.

### Profiles (what a user may do)

| Profile | For whom | Can do |
|---|---|---|
| Self-Service | Ordinary users (direction01, employe01) | Open and follow their own tickets, approve solutions. No configuration access. |
| Technician | IT staff (it01) | Handle tickets, view and edit assets. Cannot assign a ticket to a group (observed in the lab). |
| Super-Admin | Platform administrator | Everything, including configuration. |

- **Authorizations** (Habilitations) = profile + entity (+ recursive). A user can have several.
- **Dynamic profile via rules**: at import, an authorization rule gave Self-Service automatically. Technician was added to it01 by hand.
- **GLPI groups** (e.g. IT) are used to assign tickets to a team. They are separate from AD groups unless synchronised.

### Entities vs Locations — a classic confusion

| | Entity | Location |
|---|---|---|
| What it is | An **administrative / visibility** boundary (who can see what) | A **physical** place (where the thing is) |
| Effect | Controls rights and data isolation | Information only, no effect on rights |

| | | |
|---|---|---|
| Lab | Root > Site A > Direction, Employés, IT, Serveurs | Site A > Bureau Direction, Open space Employés, Local IT, Salle serveurs, Baie réseau |

### Inventory

- **GLPI Agent**: a small program installed on a computer that sends its hardware and software inventory to GLPI automatically (the server URL is set during install; a first inventory can be forced at `http://localhost:62354`).
- **Manual declaration**: used for machines without an agent and for network devices (switch, router, firewall).
- Each asset should be linked to a **user**, a **location** and a **technician / group** in charge.

## 8. ITIL ticket lifecycle

**ITIL** is a set of IT service-management best practices. GLPI follows its vocabulary.

New → Processing (assigned) → Pending (optional) → Solved → Closed

| Term | Meaning |
|---|---|
| Incident | Something is broken or degraded ("I can't reach the intranet"). |
| Request | Asking for something new ("I need software installed"). |
| Requester / Assigned to / Observer | The ticket's actors: who asked, who works on it, who is kept informed. |
| Urgency × Impact → Priority | How fast the requester needs it × how many people are affected = the order of processing. |
| ITIL category | Classification (e.g. Network > Access to an internal service), used for routing and statistics. Managed at `/front/itilcategory.php`. |
| Follow-up / Task | A message in the conversation / a technical action with time spent. |
| **Solved vs Closed** | Solved = the technician says it is fixed and documents the solution. Closed = the requester approves (or a delay expires). Until it is closed, the requester can refuse the solution and the ticket reopens. |
| History | An automatic audit trail of every change: proof of the lifecycle. |

> **Lab run:** employe01 created an incident (Network category, item CLIENT01) → the supervisor assigned it to it01 and the IT group → it01 added a diagnosis and a solution → Solved → employe01 approved → Closed.

## 9. Security principles applied

- **Least privilege**: svc-glpi can only read the directory; it is not a Domain Admin. If GLPI were compromised, the attacker would not own the domain.
- **Separation of duties**: the technician works the tickets, and a supervisor handles group assignment and configuration.
- **No secrets in documentation**: passwords are typed at prompts (`Read-Host -AsSecureString`, `ldapsearch -W`) and never captured in screenshots or reports.

- **Change safety**: a snapshot before every change; a full clone instead of modifying a VM used by another class.
- **Encryption**: LDAP on port 389 sends data in clear text; production should use **LDAPS (636)** with a certificate.
- **Root SSH login disabled**: log in as a normal user, then `su -`.

## 10. Troubleshooting method

Always go from the bottom layer up. Each step depends on the one before it:

1 Network (cable / adapter) → 2 IP / subnet → 3 DNS → 4 AD service → 5 LDAP (port + bind) → 6 GLPI settings

| Symptom met | Cause | Fix |
|---|---|---|
| `Get-ADDomain` fails, interfaces down | VirtualBox adapter "not attached" / cable disconnected | Attach to host-only, tick "Cable connected" |
| AD and GLPI cannot ping each other | Different subnets, no router | Same subnet 192.168.56.0/24 |
| `nslookup` returns two addresses | Old network card registering a stale A record | Disable the card, delete the old records |
| GLPI resolves nothing in exemple.lan (NXDOMAIN) | dhcpcd rewrote resolv.conf | `nohook resolv.conf`, restart the interface |
| GLPI "item not found" | Wrong URL (/glpi) | Use the server root |
| Copy/paste broken in RDP | Conflict between the VirtualBox shared clipboard and rdpclip | Disable the VirtualBox clipboard, sign out, reconnect |
| RDP session kicked out | The same user logged in at the console | One session per user: use RDP only |
| Root SSH "permission denied" | Root login disabled (good practice) | Log in as stagiaire, then `su -` |

## 11. Command cheat sheet

### Windows Server / PowerShell

```
ipconfig /all                                   # IP, DNS servers, adapters
Get-ADDomain                                    # is the domain reachable?
Get-ADOrganizationalUnit -Filter * | Select Name, DistinguishedName
New-ADOrganizationalUnit -Name "IT" -Path "OU=Utilisateurs,DC=exemple,DC=lan"
New-ADGroup -Name "GRP-IT" -GroupScope Global -GroupCategory Security -Path
"OU=Groupes,DC=exemple,DC=lan"
New-ADUser -Name it01 -SamAccountName it01 -Path "OU=IT,OU=Utilisateurs,DC=exemple,DC=lan" `
  -AccountPassword (Read-Host -AsSecureString "Password") -Enabled $true
Add-ADGroupMember -Identity "GRP-IT" -Members it01
Get-ADPrincipalGroupMembership svc-glpi | Select Name   # check it is NOT an admin
Get-DnsServerResourceRecord -ZoneName exemple.lan -RRType A
Add-DnsServerResourceRecordA -ZoneName exemple.lan -Name glpi01 -IPv4Address 192.168.56.20 -
CreatePtr
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses 127.0.0.1
Disable-NetAdapter -Name "Ethernet 2"
Resolve-DnsName glpi01.exemple.lan
```

### Debian

```
ip a ; ip r                                     # addresses and routes
cat /etc/network/interfaces ; cat /etc/resolv.conf
ifdown enp0s8 && ifup enp0s8                    # re-apply interface config
hostnamectl set-hostname glpi01
ping -c 3 192.168.56.10 ; nslookup dc01.exemple.lan
systemctl status apache2 mariadb ssh
php -m | grep -i ldap                           # PHP LDAP module present?
apt install ldap-utils                          # provides ldapsearch
```

### VirtualBox (host)

```
VBoxManage list vms ; VBoxManage list runningvms
VBoxManage showvminfo "Serveur_TP" | findstr NIC
VBoxManage snapshot "Serveur_TP" take "before-stage3"
```

## 12. Glossary of acronyms

| Acronym | Stands for | In one line |
|---|---|---|
| AD / AD DS | Active Directory (Domain Services) | Microsoft's directory of users, groups and computers |
| AIS | Administrateur d'Infrastructures Sécurisées | The French professional qualification this lab belongs to |
| CN | Common Name | The name part of a DN (CN=svc-glpi) |
| DC | Domain Controller / Domain Component | The AD server / a part of the domain name in a DN (DC=lan) |
| DHCP | Dynamic Host Configuration Protocol | Hands out IP settings automatically |

| | | |
|---|---|---|
| DN | Distinguished Name | The full path of an object in the directory |
| DNS | Domain Name System | Translates names into IP addresses |
| GLPI | Gestionnaire Libre de Parc Informatique | Open-source ITSM and asset management tool |
| GPO | Group Policy Object | Settings pushed to the objects in an OU |
| GUID | Globally Unique Identifier | A permanent unique ID (objectGUID) |
| ITIL | Information Technology Infrastructure Library | IT service-management best practices |
| ITSM | IT Service Management | Running IT as a service: tickets, assets, SLAs |
| LDAP / LDAPS | Lightweight Directory Access Protocol (over SSL/TLS) | Protocol to query a directory; ports 389 / 636 |
| MAC | Media Access Control address | The hardware address of a network card |
| NAT | Network Address Translation | Shares one address to reach the internet |
| NXDOMAIN | Non-Existent Domain | DNS answer: "this name does not exist" |
| OU | Organizational Unit | A folder inside AD |
| PTR | Pointer record | Reverse DNS: IP → name |
| RDP | Remote Desktop Protocol | Windows remote desktop, port 3389 |
| SLA | Service Level Agreement | Agreed response / resolution times |
| SRV | Service record | DNS record that locates a service (e.g. the DCs) |
| SSH | Secure Shell | Encrypted remote command line, port 22 |
| TLS / SSL | Transport Layer Security / Secure Sockets Layer | Encryption for network traffic |
| VLAN | Virtual LAN | Splits one physical network into isolated networks |
| VM | Virtual Machine | A computer emulated in software |

## 13. Lab access: start, connect, check, stop

Everything runs on the laptop in VirtualBox. No password is written in this guide: you type each one at its prompt.

| VM name in VirtualBox | Role | Address | How you connect | Account |
|---|---|---|---|---|
| `Serveur_TP` **(the name ends with a space)** | DC01 — Windows Server 2022, AD DS, DNS | 192.168.56.10 | Remote desktop: `mstsc /v:192.168.56.10` | `EXEMPLE\Administrateur` |
| `GLPI-Site-A` | GLPI01 — Debian 13, Apache, MariaDB, GLPI 11 | 192.168.56.20 | SSH from PowerShell: `ssh admin-sitea@192.168.56.20` | `admin-sitea` + `sudo` (since the mission). Before the mission: `stagiaire` then `su -`. Root over SSH is refused. |
| GLPI web interface | Help desk and inventory | `http://192.168.56.20/` (server root, not /glpi) | Browser on the laptop | `glpi` with source **Base interne GLPI**; `it01`, `employe01`, `direction01` with source **AD-Site-A** |

> **Before you start:** the laptop has about 12 GB of RAM, so run only these two VMs. Close `Vm Proxmox`, `debian-tp` and any other VM, plus heavy browser tabs. Never run `debian-tp` at the same time: GLPI-Site-A is its clone (same GLPI database).

### 1 — Start the lab (domain controller first)

```
cd "C:\Program Files\Oracle\VirtualBox"
.\VBoxManage.exe list runningvms                          # what is already running?
.\VBoxManage.exe startvm "Serveur_TP " --type headless    # DC first (note the space)
#  wait 2-3 minutes: AD and DNS must be up before GLPI looks for them
.\VBoxManage.exe startvm "GLPI-Site-A" --type headless
```

`--type headless` starts the VM without a window (saves memory). Use the normal VirtualBox window instead if you want to watch it boot. "already locked by a session" just means the VM is already running.

### 2 — Connect

```
mstsc /v:192.168.56.10                # Windows: remote desktop to the DC
ssh admin-sitea@192.168.56.20         # PowerShell: shell on GLPI (accept the key the first
time)
start http://192.168.56.20/           # browser: GLPI
```

> Use remote desktop **or** the VirtualBox console for the DC, never both with the same account (one session per user: the second one kicks the first out).

## 3 — Health check in 2 minutes (bottom-up)

| Check | Where | Command | Expected |
|---|---|---|---|
| Network | Laptop | `ping 192.168.56.10` / `ping 192.168.56.20` | Replies |
| GLPI web port | Laptop | `Test-NetConnection 192.168.56.20 -Port 80` | TcpTestSucceeded : True |
| AD alive | DC (PowerShell) | `Get-Service ADWS, DNS` · `Get-ADDomain` | Running · domain exemple.lan |
| GLPI services | GLPI | `systemctl status apache2 mariadb --no-pager` | active (running) |
| DNS from GLPI | GLPI | `nslookup exemple.lan` | 192.168.56.10, answered by 192.168.56.10 |
| LDAP port from GLPI | GLPI | `timeout 3 bash -c '</dev/tcp/192.168.56.10/389' && echo OPEN` | OPEN |
| End to end | Browser | Log in as `it01`, source AD-Site-A | GLPI technician interface |

If a check fails, stop there and fix that layer first: network → IP → DNS → service/port → LDAP → GLPI.

## 4 — Snapshots (restore points)

```
.\VBoxManage.exe snapshot "GLPI-Site-A" list
.\VBoxManage.exe snapshot "Serveur_TP " list
.\VBoxManage.exe snapshot "GLPI-Site-A" take "avant-xxx"      # before any change
.\VBoxManage.exe snapshot "GLPI-Site-A" restore "avant-xxx"   # VM must be powered off
```

Known restore points: Serveur_TP `INV-avant-recolement-2026-10-06`, `avant-nettoyage-dns-2026-10-06`; GLPI-Site-A `clone-initial-2026-10-06`, `avant-SSH-UFW`. Run `list` to see all of them.

## 5 — Stop the lab cleanly (GLPI first, DC last)

```
sudo poweroff                    # on GLPI (SSH session)
Stop-Computer                    # on the DC (remote desktop PowerShell)
.\VBoxManage.exe list runningvms # nothing left = done
```

Pulling the plug on a domain controller can corrupt the AD database: always shut it down from inside.

## 14. Replay: the 8 stages step by step

Use this to redo the TP from scratch or to explain it orally. Each stage: **do** → **prove** (the screenshot you would take). Take a snapshot before stages 2, 3 and 4.

| Stage | Do | Prove |
|---|---|---|
| **1 Inventory** (change nothing) | List VMs and adapters: `VBoxManage list vms`, `VBoxManage showvminfo "Serveur_TP " \| findstr NIC`. On the DC: `ipconfig /all`, `Get-ADDomain`. On GLPI: `ip a`, `cat /etc/resolv.conf`, GLPI version. | Table of machines, adapters, IPs and faults found (adapters "not attached", two subnets) |
| **2 Network & DNS** | Attach both VMs to the host-only network (cable connected). GLPI static IP 192.168.56.20 in `/etc/network/interfaces`; DNS = 192.168.56.10 in `/etc/resolv.conf`, protected with `nohook resolv.conf` in `/etc/dhcpcd.conf`. On the DC: delete stale A records, DNS client = itself, `Add-DnsServerResourceRecordA -ZoneName exemple.lan -Name glpi01 -IPv4Address 192.168.56.20`. | Ping both ways; `nslookup glpi01.exemple.lan` and `nslookup exemple.lan` from GLPI; GLPI login page in the browser |
| **3 Active Directory** | OUs `Utilisateurs` (Direction, Employes, IT), `Groupes`, `Postes`, `ComptesService`; groups `GRP-Direction`, `GRP-Employes`, `GRP-IT`, `GRP-GLPI-SVC`; users `direction01`, `employe01`, `it01`; service account `svc-glpi` in ComptesService (password typed with `Read-Host -AsSecureString`). | `Get-ADPrincipalGroupMembership svc-glpi`: no admin group |
| **4 LDAP link** | Pre-test from GLPI: `php -m \| grep -i ldap`, port 389 open, then `ldapsearch -x -H ldap://192.168.56.10 -D "svc-glpi@exemple.lan" -W -b "DC=exemple,DC=lan" "(sAMAccountName=it01)" cn`<br>GLPI: Configuration › Authentication › LDAP directories › add **AD-Site-A**: server 192.168.56.10, port 389, Base DN `DC=exemple,DC=lan`, bind DN `CN=svc-glpi,OU=ComptesService,DC=exemple,DC=lan`, filter `(&(objectClass=user)(objectCategory=person))`, login field `samaccountname`, sync field `objectguid`. Disable the old `AD_Ecole`. Import direction01, employe01, it01. | "Test" tab all green; imported users list; it01 logs in with source AD-Site-A |
| **5 Profiles** | Self-Service for the requesters (automatic rule at import); add Technician to it01; GLPI group IT with it01. | employe01 is refused the configuration pages; it01 sees the technician interface |
| **6 Inventory** | Entities (Site A › Direction, Employés, IT, Serveurs) and locations; computers DC01 (agent), GLPI01 and CLIENT01 (manual); switch, router and firewall declared manually. | Computers list and network devices list |

| | | |
|---|---|---|
| **7 Ticket** | employe01 opens an incident (category Network, item CLIENT01) → assign to it01 + group IT → follow-up → solution → Solved → employe01 approves → Closed. | Ticket history showing the full lifecycle |
| **8 Acceptance tests** | Rerun the health check of the previous section and the test plan; note any deviation and its fix. | Completed test table, all "OK" |

Companion to the French lab report in `rapport/`. Real values come from the home lab; no passwords are recorded.